Alessandro **Sanvito**

SOFTWARE AND AI ENGINEER

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Experience

Optiver *Amsterdam, Netherlands*

SOFTWARE ENGINEER *August 2023 - Current*

- On a systematic options trading desk, took new asset classes from nothing to a working baseline systematic strategy — including data sourcing, cleaning, and validation — that quantitative researchers then iterate on.
- Exercised the research judgment in that bridge: choosing plausible signals for each new asset class and defining the validation and acceptance criteria a baseline strategy must meet before it is handed off.
- Broadened that work with infrastructure: substantially reduced backtest runtime and moved ML model training from local to cloud, working across Python, Spark, Polars, DuckDB, PostgreSQL, and Delta Lake.

Mercedes-Benz (AI-SEE Project) *Stuttgart, Germany*

AI RESEARCH INTERN AND MASTER THESIS STUDENT *May 2022 - June 2023*

- Conducted research on 3D human avatar modeling from monocular video data in diverse real-world environments.
- Designed and implemented generative neural models, including NeRFs and Diffusion Models, to improve computer vision performance in adverse weather conditions.
- Collaborated with a cross-functional R&D team, resulting in a first-author publication at IEEE IV 2024 and a co-authored publication at ICCV 2023.

Education

KTH Royal Institute of Technology *Stockholm, Sweden*

MSC. IN ICT INNOVATION, DATA SCIENCE *Sept. 2020 - May 2023*

- Final grade: A - Excellent
- Implemented from scratch a diverse range of neural network architectures in NumPy, including MLPs, Hopfield Networks, and Deep Belief Networks.
- Reproduced data mining-related papers in Python and Java.

Polytechnic University of Milan *Milan, Italy*

MSC. IN COMPUTER SCIENCE AND ENGINEERING *Sept. 2020 - Jul. 2023*

- **1st place (academic)** at the international ACM RecSys Challenge 2021: co-developed the winning ML solution on the Twitter engagement dataset with the university team under prof. Paolo Cremonesi, using Dask, CatBoost, and XGBoost.
- Final grade: 110/110 cum laude
- Built a recommender system for a book catalogue with NumPy, SciPy, and Pandas.

Polytechnic University of Milan *Milan, Italy*

BSC. IN ENGINEERING OF COMPUTING SYSTEMS *Sept. 2017 - Jul. 2020*

- Final grade: 109/110
- Created a performance-oriented graph manipulation tool in C.

Publications

2024	**HINT: Learning Complete Human Neural Representations from Limited Viewpoints**, **Sanvito Alessandro** (first author), Ramazzina Andrea, Walz Stefanie, Bijelic Mario, Heide Felix	*IEEE IV 2024*
2023	**ScatterNeRF: Seeing Through Fog with Physically-Based Inverse Neural Renderings**, Ramazzina Andrea, Bijelic Mario, Walz Stefanie, **Sanvito Alessandro**, Scheuble Dominik, Heide Felix	*ICCV 2023*
2022	**United We Stand, Divided We Fall: Leveraging Ensembles of Recommenders to Compete with Budget Constrained Resources**, Maldini Pietro, **Sanvito Alessandro**, Surricchio Mattia	*ACM recsys'22*
2021	**Lightweight and Scalable Model for Tweet Engagements Predictions in a Resource-constrained Environment**, Carminati Luca, Lodigiani Giacomo, Maldini Pietro, Meta Samuele, Metaj Stiven, Pisa Arcangelo, **Sanvito Alessandro**, Surricchio Mattia, Maurera Fernando B. Peréz, Bernardis Cesare, Ferrari Dacrema Maurizio	*ACM recsys'21*

Skills

Programming	Python, SQL, Java, C++, C
Research & ML	PyTorch, generative models (NeRFs, Diffusion), computer vision, recommender systems, gradient boosting (CatBoost, XGBoost), scikit-learn
Data Engineering	Spark, Polars, DuckDB, PostgreSQL, Delta Lake, Pandas, NumPy, MLlib, cloud compute
Tools & Platforms	Git, Docker, Linux, CI/CD, pytest